The solution key includes all the extra questions not included in the EEE4773 midterm exam.

1. (12 points) **Consider a set of i.i.d. samples $\{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}$, and its corresponding target labels $\{t_i\}_{i=1}^N$, where $t_i \in \mathbb{R}$. Answer the following questions:**

   (a) (3 points) **Write down the mapper function for a linear regression model with M-degree polynomial features.**

   The mapper function for a linear regression model with M-degree polynomial features is:

$$y(x_i) = \sum_{j=0}^{M} w_j x_i^j$$

   (b) (3 points) **Consider the Mean Absolute Error (MAE) objective function with a Lasso regularizer term. Write down this objective function.**

   The Mean Absolute Error (MAE) objective function with a Lasso regularizer term is:

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^{N} |t_i - y(x_i)| + \lambda \sum_{j=0}^{M} |w_j| \\ &= \|\mathbf{t} - \mathbf{y}\|_1 + \lambda \|\mathbf{w}\|_1 \end{aligned}$$

   (c) (3 points) **How would you explain the objective function you wrote down in Bayesian terms?**

   We observe that:

$$\begin{aligned} \arg_{\mathbf{w}} \min J(\mathbf{w}) &= \arg_{\mathbf{w}} \max \exp(-J(\mathbf{w})) \\ &= \arg_{\mathbf{w}} \max \exp\left(-\sum_{i=1}^{N} |t_i - y(x_i)| - \lambda \sum_{j=0}^{M} |w_j|\right) \\ &= \arg_{\mathbf{w}} \max \exp\left(-\sum_{i=1}^{N} |t_i - y(x_i)|\right) \exp\left(-\lambda \sum_{j=0}^{M} |w_j|\right) \\ &= \arg_{\mathbf{w}} \max \prod_{i=1}^{N} \exp\left(-|t_i - y(x_i)|\right) \prod_{j=0}^{M} \exp\left(-\lambda |w_j|\right) \\ &= \arg_{\mathbf{w}} \max \mathcal{L}(\mathbf{t}|\mathbf{y}, 1) \mathcal{L}(\mathbf{w}|0, 1/\lambda) \\ &= \arg_{\mathbf{w}} \max P(\mathbf{t}|\mathbf{w}) P(\mathbf{w}) \\ &\propto \arg_{\mathbf{w}} \max P(\mathbf{w}|\mathbf{t}) \end{aligned}$$

In other words, the objective function assumes a Laplacian distribution on the target variables, $\mathbf{t}$, centered at the estimations, $\mathbf{y}$, with scale 1, with a prior belief on the parameter values $\mathbf{w}$. The prior is also Laplacian-distributed on the parameters $\mathbf{w}$, centered at 0 with scale $1/\lambda$. Thus, this objective function solves for the MAP solution of $\mathbf{w}$.

(d) (3 points) **Does this formulation exhibit a conjugate prior relationship? Why or why not?**

No. The product of two Laplacian distributions cannot be simplified to parametrically resemble another Laplacian. Thus the resulting posterior does not the same parametric shape as the prior (Laplacian), therefore this formulation does not exhibit a conjugate prior.

2. (10 points) **Consider training a linear regression model, $y(\mathbf{x}) = \mathbf{w}^T\phi(\mathbf{x}) + w_0$, with the objective function of Mean Squared Error (MSE). Answer the following questions:**

(a) (5 points) **Which parameter estimation method, Maximum Likelihood Estimation (MLE) or Maximum A Posteriori (MAP), is more effective in preventing overfitting? Explain why and specify under what conditions this advantage holds.**

MAP is more effective at preventing overfitting in the case of a small sample size. We know that MAP assumes a prior on the parameters of interests which, effectively, regularizes them to control overfitting. The advantage of using MAP with small sample sizes holds if the prior belief on the parameters is encoded to include its true unknown value.

As the sample size increases, and provided MAP encodes an adequate prior, both MLE and MAP converge to the same solution.

(b) (5 points) **Discuss how cross-validation mitigates overfitting and supports reliable estimation of model performance.**

Cross-validation mitigates overfitting by evaluating a model on multiple independent subsets of data rather than a single train-test split. By partitioning the dataset into k folds and iteratively training on k-1 folds while validating on the remaining fold, cross-validation ensures the model's performance is not inflated by memorizing one particular test set. This approach provides a more robust estimate of generalization performance by averaging results across all folds, revealing whether the model truly learns underlying patterns or simply fits to noise.

Additionally, comparing training and validation scores across folds helps identify overfitting. If training performance significantly exceeds validation performance, the model is likely too complex for the available data. This systematic validation process guides hyperparameter tuning and model selection decisions based on reliable performance metrics rather than potentially misleading single-split results.

3. (15 points) **Suppose we have a dataset $\{x_i\}_{i=1}^N$, where each $x_i \geq 0$ denotes the observed time-to-failure of a transistor. Assume the samples are independent**

and identically distributed (i.i.d.), and each sample is drawn from a Weibull random variable with probability density function:

$$\begin{aligned} P(x|\lambda, k) &= \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp^{-(x/\lambda)^k} \\ &= k\lambda^{-1}x^{k-1}\lambda^{-(k-1)} \exp\left(-x^k\lambda^{-k}\right) \end{aligned}$$

where $\lambda, k > 0$**.**

Moreover, consider the Inverse Gamma probability density function as the prior probability on the hyperparameter λ**,**

$$P(\lambda|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-\alpha-1} \exp\left(-\frac{\beta}{\lambda}\right)$$

where $\alpha, \beta > 0$**.**

Answer the following questions:

(a) (5 points) **Derive the maximum likelihood estimate (MLE) for the parameter** λ**. Show your work.**

The observed data likelihood is:

$$\begin{aligned} \mathcal{L}^0 &= \prod_{i=1}^{N} k\lambda^{-1}x_i^{k-1}\lambda^{-(k-1)} \exp\left(-x_i^k\lambda^{-k}\right) \\ &= \prod_{i=1}^{N} k\lambda^{-k}x_i^{k-1} \exp\left(-x_i^k\lambda^{-k}\right) \end{aligned}$$

The log-likelihood is given by:

$$\begin{aligned} \mathcal{L} &= \ln \mathcal{L}^0 \\ &= \sum_{i=1}^{N} \left(\ln(k) - k\ln(\lambda) + (k-1)\ln(x_i) - x_i^k\lambda^{-k}\right) \end{aligned}$$

We can now find the MLE estimation for β:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} = 0 &\iff \sum_{i=1}^{N}\left(-\frac{k}{\lambda} + \frac{kx_i^k}{\lambda^{k+1}}\right) = 0 \iff \sum_{i=1}^{N}\left(-k + \frac{kx_i^k}{\lambda^k}\right) = 0 \\ &\iff \sum_{i=1}^{N}\left(-1 + \frac{x_i^k}{\lambda^k}\right) = 0 \iff N = \frac{1}{\lambda^k}\sum_{i=1}^{N} x_i^k \\ &\iff \lambda^k = \frac{1}{N}\sum_{i=1}^{N} x_i^k \iff \lambda = \left(\frac{1}{N}\sum_{i=1}^{N} x_i^k\right)^{1/k} \end{aligned}$$

(b) (5 points) **Consider $k = 1$. Derive the maximum a posteriori (MAP) estimate for the parameter λ. Show your work.**

The observed data likelihood is:

$$\begin{aligned}
\mathcal{L}^0 &= \left(\prod_{i=1}^{N} k\lambda^{-k} x_i^{k-1} \exp\left(-x_i^k \lambda^{-k}\right)\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-\alpha-1} \exp\left(-\frac{\beta}{\lambda}\right) \\
&= \left(\prod_{i=1}^{N} \lambda^{-1} \exp\left(-x_i \lambda^{-1}\right)\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-\alpha-1} \exp\left(-\frac{\beta}{\lambda}\right) \\
&= \lambda^{-N} \left(\prod_{i=1}^{N} \exp(-x_i \lambda^{-1})\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-\alpha-1} \exp\left(-\frac{\beta}{\lambda}\right) \\
&= \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-N-\alpha-1} \exp(-\beta\lambda^{-1}) \left(\prod_{i=1}^{N} \exp(-x_i \lambda^{-1})\right)
\end{aligned}$$

The log-likelihood is given by:

$$\begin{aligned}
\mathcal{L} &= \ln \mathcal{L}^0 \\
&= \alpha \ln(\beta) - \ln(\Gamma(\alpha)) - (N+\alpha+1)\ln(\lambda) - \beta\lambda^{-1} - \sum_{i=1}^{N} x_i \lambda^{-1}
\end{aligned}$$

We can now find the MAP estimation for β:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \lambda} = 0 &\iff -(N+\alpha+1)\frac{1}{\lambda} - \beta\lambda^{-2} + \sum_{i=1}^{N} x_i \lambda^{-2} = 0 \\
&\iff -(N+\alpha+1) + \frac{1}{\lambda}\left(-\beta + \sum_{i=1}^{N} x_i\right) = 0 \\
&\iff \frac{1}{\lambda}\left(-\beta + \sum_{i=1}^{N} x_i\right) = (N+\alpha+1) \\
&\iff \lambda = \frac{\sum_{i=1}^{N} x_i - \beta}{N+\alpha+1}
\end{aligned}$$

(c) (5 points) **For the case of $k = 1$, show that the Weibull-Inverse Gamma distribution is a conjugate prior for the parameter λ, and derive the corresponding update equations for the hyperparameters α and β.**

From the previous question, we found that the observed data likelihood can be simplified to:

$$\begin{aligned}
\mathcal{L}^0 &= \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-N-\alpha-1}\exp(-\beta\lambda^{-1})\left(\prod_{i=1}^{N}\exp(-x_i\lambda^{-1})\right)\\
&= \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-N-\alpha-1}\exp(-\beta\lambda^{-1})\exp\left(\sum_{i=1}^{N}-x_i\lambda^{-1}\right)\\
&= \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-N-\alpha-1}\exp\left(-\left(\beta+\sum_{i=1}^{N}x_i\right)\lambda^{-1}\right)
\end{aligned}$$

We find that this expression is parametrically equivalent to the prior distribution (by a constant factor), where $\alpha \longrightarrow \alpha + N$ and $\beta \longrightarrow \beta + \sum_{i=1}^{N} x_i$. Therefore, the case of $k = 1$, the Weibull-Inverse Gamma distribution is a conjugate prior for the parameter λ.

4. (10 points) **Consider a labeled training dataset $\{(\mathbf{x}_i, t_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^D$ and $t_i \in \mathbb{R}$ $\forall i$. Let $y(x_i) = \mathbf{w}^T\phi(\mathbf{x}_i)$ be the mapper function. Answer the following questions:**

   (a) (5 points) **The solution for w is the one that maximizes this weighted likelihood:**

   $$\mathbf{w}^* = \arg_{\mathbf{w}}\max G(\mathbf{t}-\mathbf{y}|\mu = 0, \sigma^2 = 1)G(\mathbf{w}|\mu_{\mathbf{w}} = 0, \sigma^2_{\mathbf{w}} = \alpha)$$

   Without taking derivatives, provide the analytical solution for $\mathbf{w}^*$ and briefly explain why.

   The solution for $\mathbf{w}$ that maximizes this expression, is the one that assumes a data likelihood on the residuals $\epsilon = \mathbf{t} - \mathbf{y}$ to be a Gaussian distribution with mean 0 and variance 1, and a prior probability on the parameters $\mathbf{w}$, also Gaussian-distributed with mean 0 and variance α. Thus, this is a Least squares solution with the Ridge regularizer, where the regularization hyperparameter is defined as $\frac{1}{\alpha}$. The solution for $\mathbf{w}$ is thus $\mathbf{w} = \left(\mathbf{X}^T\mathbf{X} + \frac{1}{\alpha}\mathbf{I}\right)^{-1}\mathbf{X}\mathbf{t}$, where $\mathbf{X}$ is the data matrix of shape $N \times D$, and $\mathbf{t}$ is the labels vector of shape $N \times 1$.

   (b) (5 points) **Explain the role of α in the solution for w. Describe what happens to the solution in the limits $\alpha \longrightarrow 0$ and $\alpha \longrightarrow \infty$.**

   The hyperparameter α controls the trade-off between avoiding overfitting and goodness-of-fit. As $\alpha \to 0$, the solution for $\mathbf{w}$ will become more regularized, and the solution for $\mathbf{w}$ will converge to the zeros-vector. As $\alpha \to \infty$, the solution for $\mathbf{w}$ will effectively have no regularizer, and the solution for $\mathbf{w}$ will converge to the MLE solution $\mathbf{W} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}\mathbf{t}$.

5. (15 points) **A simple email classifier uses two binary features:**

   - $x_1 = 1$ **if the email contains the word “discount”, otherwise 0.**
   - $x_2 = 1$ **if the email contains the word “meeting”, otherwise 0.**

Class (c)	$P(t=c)$	Count($x_1=1$)	Count($x_2=1$)	Total Samples
Spam ($t=1$)	0.4	6	2	10
Not Spam ($t=0$)	0.6	2	7	10

You have the following training data statistics:

Answer the following questions:

(a) (5 points) **Compute the conditional probabilities:** $P(x_1=1|t=1)$**,** $P(X_2=1|t=1)$**,** $P(x_1=1|t=0)$**, and** $P(x_2=1|t=0)$**.**

From the provided values, we find:

$$\begin{aligned} P(x_1=1|t=1) &= \frac{6}{10} \\ P(X_2=1|t=1) &= \frac{2}{10} \\ P(x_1=1|t=0) &= \frac{2}{10} \\ P(x_2=1|t=0) &= \frac{7}{10} \end{aligned}$$

(b) (8 points) **Compute the posterior probabilities for a new email with**

$$\mathbf{x} = [x_1=1, x_2=0]$$

The posterior probability is defined as:

$$P(t|\mathbf{x}=[x_1=1,x_2=0]) = \frac{P(\mathbf{x}=[x_1=1,x_2=0]\cap t)P(t)}{P(\mathbf{x}=[x_1=1,x_2=0])}$$

For $t=0$,

$$\begin{aligned} P(\mathbf{x}=[x_1=1,x_2=0]\cap t=0) &= P(t=0)P(x_1=1|t=0)P(x_2=0|t=0) \\ &= 0.6\times\frac{2}{10}\times\frac{3}{10} \\ &= \frac{36}{1000} = 0.036 \end{aligned}$$

For $t=1$,

$$\begin{aligned} P(\mathbf{x}=[x_1=1,x_2=0]\cap t=1) &= P(t=1)P(x_1=1|t=1)P(x_2=0|t=1) \\ &= 0.4\times\frac{6}{10}\times\frac{8}{10} \\ &= \frac{192}{1000} = 0.192 \end{aligned}$$

Lastly,

$$\begin{aligned} P(\mathbf{x}=[x_1=1,x_2=0]) &= P(\mathbf{x}=[x_1=1,x_2=0]\cap t=0)P(t=0) + P(\mathbf{x}=[x_1=1,x_2=0]\cap t= \\ &= 0.036\times 0.6 + 0.192\times 0.4 \\ &\approx 0.0984 \end{aligned}$$

With this, we find,

$$P(t=0|\mathbf{x}=[x_1=1,x_2=0]) = \frac{0.036\times 0.6}{0.0984} \approx 0.2195$$
$$P(t=1|\mathbf{x}=[x_1=1,x_2=0]) = \frac{0.192\times 0.4}{0.0984} \approx 0.7805$$

(c) (2 points) **Which class would Naive Bayes predict for this email?**

Based on the calculations from the previous question, the email with $[x_1=1,x_2=0]$ would be classified as class 1, as it has the largest probability of assignment (0.7805).

6. (15 points) **Consider the dataset $\mathbf{X}=\{x_i\}_{i=1}^N$, where $x_i\in\mathbb{N}_0,\forall i$. Suppose your goal is to perform density estimation using a Mixture Model, in particular, a Poisson Mixture Model. Its data likelihood can be written as:**

$$P(x)=\sum_{k=1}^K \pi_k p_k\left(x|\lambda_k\right)$$

where

$$\sum_{k=1}^K \pi_k = 1 \quad \text{and} \quad 0\le\pi_k\le 1$$

and

$$p_k\left(x|\lambda_k\right)=\frac{\lambda_k^x e^{-\lambda_k}}{x!}$$

with $\lambda_k>0 \quad \forall k$. Answer the following questions:

(a) (2 points) **Assuming your data is i.i.d., write down the observed data likelihood, $\mathcal{L}^0$.**

The observed data likelihood is:

$$\mathcal{L}^0=\prod_{i=1}^N\sum_{k=1}^K \pi_k\frac{\lambda_k^{x_i}e^{-\lambda_k}}{x_i!}$$

(b) (2 points) **Use the Expectation-Maximization (EM) algorithm to find the maximum likelihood solutions. Start by introducing the hidden latent variable Z. Describe precisely what they are.**

The hidden latent variables z_i are defined as the Poisson component from which sample x_i was drawn from, $\forall i$. Moreover, since we have a total of K Poisson components, $z_i\in\{1,2,\ldots,K\}$.

(c) (2 points) **For the hidden variable you defined above, write down the complete data likelihood, $\mathcal{L}^c$.**

The complete data likelihood is given by:

$$\mathcal{L}^c = \prod_{i=1}^{N} \pi_{z_i} \frac{\lambda_{z_i}^{x_i} e^{-\lambda_{z_i}}}{x_i!}$$

(d) (4 points) **Write down the EM optimization function, $Q(\Theta, \Theta^t)$, where $\Theta = \{\pi_k, \lambda_k\}_{k=1}^{K}$. Your final solution should contain the sum of simple (natural-)log-terms.**

The EM optimization function is given by:

$$\begin{aligned}
Q(\Theta, \Theta^t) &= \mathbb{E}_z[\ln(\mathbf{L}^c)|\mathbf{X}, \Theta^t] \\
&= \sum_{z_i=1}^{K} \ln(\mathbf{L}^c) P(z_i|x_i, \Theta^t) \\
&= \sum_{k=1}^{K} \left(\sum_{i=1}^{N} \ln(\pi_k) + x_i \ln(\lambda_k) - \lambda_k - \ln(x_i!) \right) P(z_i = k|x_i, \Theta^t) \\
&= \sum_{k=1}^{K} \left(\sum_{i=1}^{N} \ln(\pi_k) + x_i \ln(\lambda_k) - \lambda_k - \ln(x_i!) \right) C_{ik}
\end{aligned}$$

(e) (5 points) **Derive the update equations for the parameters λ_k.** The solution for the parameter σ_k is:

$$\begin{aligned}
\frac{\partial Q(\Theta, \Theta^t)}{\partial \lambda_k} &= 0 \\
\sum_{i=1}^{N} \left(\frac{x_i}{\lambda_k} - 1 \right) C_{ik} &= 0 \\
\frac{1}{\lambda_k} \sum_{i=1}^{N} x_i C_{ik} - \sum_{i=1}^{N} C_{ik} &= 0 \\
\lambda_k &= \frac{\sum_{i=1}^{N} x_i C_{ik}}{\sum_{i=1}^{N} C_{ik}}
\end{aligned}$$

7. (15 points) **Consider a dataset with N samples $\{\mathbf{x}_n\}_{n=1}^{N}$ where each $\mathbf{x}_n \in \mathbb{R}^D$. Suppose you use the K-Means algorithm with a Euclidean distance metric to cluster the data into K components. In other words, your objective function is**

$$J(\Theta, U) = \sum_{n=1}^{N} \sum_{k=1}^{K} u_{nk} \|\mathbf{x}_n - \theta_k\|_2^2 \quad \text{such that } u_{nk} \in \{0, 1\} \text{ and } \sum_{k=1}^{K} u_{nk} = 1$$

where u_{nk} are cluster assignments, θ_k is the k^{th} cluster representative.

(a) (3 points) **Provide the pseudo-code for K-Means.**

The pseudo-code for K-Means is as follows:

1. Set number of clusters, K, and the distance metric.
2. Initialize the cluster centroids, $\{\theta_k\}_{k=1}^{K}$.
3. For each point, compute the distance to the cluster centroid. Assign membership to the label corresponding to the closest cluster centroid, $\{u_n\}_{n=1}^{N}$.
4. Update cluster center, θ_k, to the mean of points in cluster $\theta_k = \frac{\sum_{x_n \in C_k} x_n}{N_k}$.
5. Go back to step 3 and repeat until convergence criteria is met. Convergence criteria includes reaching maximum number of iterations or change in the cluster centroids and/or membership values is small.

(b) (5 points) **Optimize the objective function for the cluster representatives θ_k.**

The objective function is as follows:

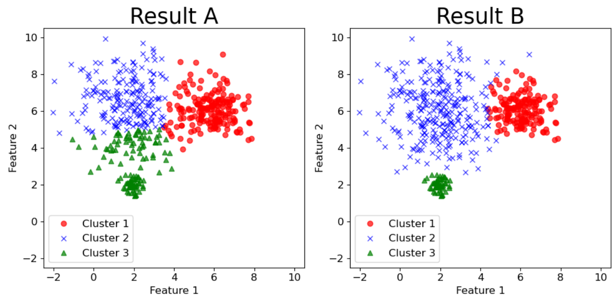
$$J(\Theta, U) = \sum_{n=1}^{N}\sum_{k=1}^{K} u_{nk}\|\mathbf{x}_n - \theta_k\|_2^2 = \sum_{n=1}^{N}\sum_{k=1}^{K} u_{nk}(\mathbf{x}_n - \theta_k)^T(\mathbf{x}_n - \theta_k)$$

$$\text{such that } u_{nk} \in \{0,1\} \text{ and } \sum_{k=1}^{K} u_{nk} = 1$$

$$\begin{aligned}
\frac{\partial J(\Theta, U)}{\partial \theta_k} &= 0 \\
\sum_{n=1}^{N} -2u_{nk}(\mathbf{x}_n - \theta_k)^T &= 0 \\
\sum_{n=1}^{N} u_{nk}x_n &= \sum_{n=1}^{N} u_{nk}\theta_k \\
\theta_k &= \frac{\sum_{n=1}^{N} u_{nk}x_n}{\sum_{n=1}^{N} u_{nk}} \\
\theta_k &= \frac{\sum_{x_n \in C_k} x_n}{N_k}
\end{aligned}$$

(c) (5 points) **In the figure below, which plot displays clusters generated by K-Means and which plot displays clusters generated by a Gaussian Mixture Model? Explain your reasoning.**

Result A was generated using K-Means clustering, which partitions the data by minimizing the distance between each point and its assigned cluster center. In contrast, Result B shows the best-fitting distribution of the data, which is more accurately captured by GMMs.



(d) (2 points) **K-means is sure to converge, but this result may be a local optimum rather than the global solution. Explain why this occurs.**

K-means is guaranteed to converge because each iteration reduces (or keeps constant) the objective function, which is the total within-cluster sum of squared distances. In each iteration, the algorithm first assigns points to their nearest centroids, minimizing the distance for those assignments, and then updates each centroid to the mean of its assigned points, which also minimizes the total distance within that cluster. Since the objective function is non-increasing and bounded below, the process must eventually stop. However, convergence does not guarantee finding the global minimum because the algorithm's result depends on the initial placement of centroids. Different initializations can lead to different partitions of the data, and K-means can settle into a local optimum, a stable configuration where small changes would increase the objective function, even though a better global configuration might exist.

8. (8 points) **True (T) or False (F).**

T. **Mixture models can be applied for both data modeling and clustering tasks.**

F. **The EM algorithm guarantees convergence to the global optimum solution.**

F. **In MLE estimation we optimize the posterior distribution.**

T. **In Linear Regression, we can represent the data using nonlinear transformations of the input variables.**

F. **When k-fold cross-validation is used for hyperparameter tuning, overfitting will not occur.**

F. **The Naive Bayes Classifier learns a linear decision function.**

F. **K-Means uses a soft membership assignment.**

F. **Ridge regularizer allows for feature selection.**